

Midterm Exam

1. (18 pt) Solving a Linear System of Equations

Consider the system of linear equations:

$$\begin{cases} x_1 + 2x_2 - x_3 + 4x_4 = 3 \\ 2x_1 - x_2 + 3x_3 + 5x_4 = -1 \\ 3x_1 + 4x_2 + 2x_3 - x_4 = 2 \end{cases}$$

- (a) (3 pt) Write the coefficient matrix A and the corresponding augmented matrix.
- (b) (5 pt) Compute the row-reduced echelon form (RREF) of the augmented matrix.
- (c) (5 pt) Express the solution set of the system.
- (d) (3 pt) Determine the rank and nullity (i.e., dimension of null space) of the coefficient matrix A .
- (e) (2 pt) Find a basis for the null space of A . (no need to prove it is a basis)

2. (15 pt) Question Answering

- (a) (3 pt) Is $\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 + x_2 = 1\}$ a linear space? Justify your answer.
- (b) (3 pt) Suppose $u, v \in \mathbb{R}^{n \times 1}$, and $A, B, C \in \mathbb{R}^{n \times n}$. Which of the following matrix products are valid? $uAv, u^T ABv, ABCvuA, u^T Avu^T BCvv^T A$. Briefly justify your answer.
- (c) (3 pt) Suppose a matrix $A \in \mathbb{R}^{5 \times 8}$ and $\text{rank}(A) = 3$. Find $\dim(N(A))$ and $\dim(N(A^T))$. Briefly explain your calculations.
Reminder: For any matrix Z , $N(Z)$ denotes the null space of Z .
- (d) (3 pt) State a necessary and sufficient condition for a diagonal matrix $A \in \mathbb{R}^{n \times n}$ to be invertible, and provide a concise proof.
- (e) (3 pt) Suppose $A \in \mathbb{R}^{n \times n}$. Is it possible to have a matrix A where $A \neq I$ and $A \neq 0$ such that $A^2 = A$? State your answer and briefly explain why.

3. (16 pt) Matrix Transpose, Inverse and Multiplication

- (a) (4 pt) Consider the matrix

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

Find two different 3×2 matrices C (whose elements can only be 0, 1, or -1) such that $AC = I_2$, where $I_2 \in \mathbb{R}^{2 \times 2}$ is the identity matrix.

- (b) (4 pt) Let

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}.$$

Compute AB and $B^T A^T$.

- (c) (4 pt) Calculate

$$\begin{pmatrix} 2 & 5 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 0 & 0 & 3 & 4 \\ 0 & 0 & 2 & 3 \end{pmatrix}^{-1}.$$

- (d) (4 pt) Show that if $A^3 = 2I$ and $B = A^2 - 2A + 2I$, then

$$B^{-1} = \frac{1}{10}(A^2 + 3A + 4I).$$

4. (16 pt) True or False

For each question, you do NOT need to justify. (Only writing T or F is enough)

- (a) If a matrix A has full row rank, then for any vector $\mathbf{b}$, the linear system $A\mathbf{x} = \mathbf{b}$ has at most one solution.
- (b) If a square matrix is invertible, then the intersection of its column space and null space only contains one element which is the zero vector.
- (c) For any matrix A , the matrix $A^T A$ is symmetric.
- (d) The intersection of two subspaces of a certain linear space must be a subspace.
- (e) If a set of vectors in $\mathbb{R}^n$ is linearly dependent, then it cannot span $\mathbb{R}^n$.
- (f) Any set of vectors in $\mathbb{R}^n$ containing the zero vector is linearly dependent.

(g) If the columns of a matrix are linearly independent, its rows must also be linearly independent.

(h) If a homogeneous linear system of equations has more variables than equations, then the system must have a nonzero solution.

5. (15 pt) Matrix Multiplication

In the Llama2 architecture by Meta AI, a key component is defined as:

$$Z = ((XU) \circ (XV))W$$

where: $X \in \mathbb{R}^{n \times d}$, $U, V \in \mathbb{R}^{d \times k}$, $W \in \mathbb{R}^{k \times m}$. The open circle “ $\circ$ ” denotes the Hadamard (element-wise) product: $[A \circ B]_{ij} = A_{ij}B_{ij}$ for any two matrices $A, B \in \mathbb{R}^{n \times k}$.

- (a) (4 pt) If we multiply only the 3rd column of W by 2, while keeping other parts of W the same, what happens to the 3rd column of Z ? Justify your answer.
- (b) (4 pt) If we multiply only the 4th column of W by 2, while keeping other parts of W the same, what happens to the 3rd column of Z ? Justify your answer.
- (c) (4 pt) If we multiply only the 3rd row of X by 2, while keeping other parts of X the same, what happens to the 3rd row of Z ? Justify your answer.
- (d) (3 pt) If we multiply only the 4th row of X by 2, while keeping other rows the same, what happens to the 3rd row of Z ? Justify your answer.

6. (20 pt) Rank, Independence

- (a) (5 pt) Consider the elementary row operation of adding a multiple of one row to another $R_i \leftarrow cR_j + R_i$ ($i \neq j$). In the class, we mentioned the result: applying this operation to a matrix $A \in \mathbb{R}^{m \times n}$ is equivalent to left-multiplying A by the elementary matrix $E_{cR_j+R_i}$. Prove this result for the case $m = 5, i = 4, j = 2$.
- (b) (5 pt) Let $v_1, v_2, \dots, v_k$ be vectors in $\mathbb{R}^n$. Denote $w_1 = v_1 + \alpha v_2$, where $\alpha \in \mathbb{R}$. Prove: $\text{span}\{v_1, v_2, v_3, \dots, v_k\} = \text{span}\{w_1, v_2, v_3, \dots, v_k\}$.
- (c) (5 pt) Let $v_1, v_2, \dots, v_k$ be vectors in $\mathbb{R}^n$. Define $w_i = v_1 + v_2 + \dots + v_i$ for $i = 1, 2, \dots, k$. Prove that if the set $\{w_1, w_2, \dots, w_k\}$ is linearly independent, then the set $\{v_1, v_2, \dots, v_k\}$ is also linearly independent.
- (d) (5 pt) Prove the following statements: Suppose $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{l \times m}$, if $\text{rank}(A) = n$, and $\text{rank}(B) = m$, then $\text{rank}(BA) = n$.
Hint: There are multiple methods to prove it. One way is to consider the column form of A .